# Physics I (Spring 2024): Homework #1(-Part 2)

Department of Physics and Astronomy, SNU

5 problems; 50 points in total; Posted Mar. 15; Due Mar. 29, 2024

• By turning in your homework, you acknowledge that you have not received any unpermitted aid, nor have compromised your academic integrity during its preparation.

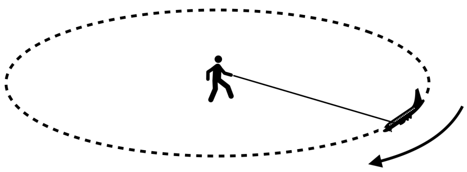
• Exhibit all intermediate steps to receive full credits. You are welcomed to use a scientific calculator — physical one or the one on your cellphone. Obtain numerical results accurate to *two* significant figures.

• Gravitational acceleration $g = 9.8\,\mathrm{m/s^2}$. Assume negligible friction and air resistance, unless stated otherwise. Assume also that any cord connecting one object to another is massless, unless stated otherwise.

1. [10 pt] A girl whirls a toy plane in uniform circular motion in a horizontal circle. At time $t_1 = 2.0$ second, the plane is at $\mathbf{x}_1 = (2.0\,\mathrm{m})\hat{\mathbf{i}} + (3.0\,\mathrm{m})\hat{\mathbf{j}}$ with velocity $\mathbf{v}_1 = (9.0\,\mathrm{m/s})\hat{\mathbf{j}}$ and acceleration in the positive $x$ direction, measured on a horizontal $x$-$y$ coordinate system. At time $t_2 = 4.0$ second, the plane has velocity $\mathbf{v}_2 = (-9.0\,\mathrm{m/s})\hat{\mathbf{i}}$ and acceleration in the positive $y$ direction. Let us assume that $t_2 - t_1$ is less than one period.

(a) What is the period of revolution, $T$, of this motion? What is the radius $R$ of the plane's circular path? What is the magnitude of the plane's centripetal acceleration?

(b) What is the $(x, y)$-position of the center of the circular path? What is the plane's position $\mathbf{x}_2$ at $t_2$?

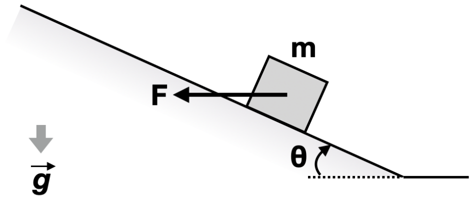


2. [10 pt] A block of mass $m = 50\,\mathrm{kg}$ is about to slide down a long, frictionless ramp that is fixed in space (with $\theta = 30°$; see figure).

(a) What is the magnitude of a horizontal force $\mathbf{F}_1$ that a worker must apply to keep the block still? What is the magnitude of the force on the block from the ramp, i.e., the normal force?

(b) Now, the block is being pushed up the ramp with a constant speed of $v = 2.4\,\mathrm{m/s}$ (measured along the slope of the ramp) by a horizontal force $\mathbf{F}_2$. What are the magnitudes of $\mathbf{F}_2$ and the force on the block from the ramp? Use a free-body diagram if needed.

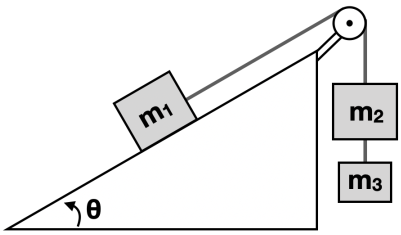
(c) Finally, the block is pushed up the ramp with a constant acceleration of $a = 3.1\,\mathrm{m/s^2}$ (measured along the slope of the ramp) by a horizontal force $\mathbf{F}_3$. What are the magnitudes of $\mathbf{F}_3$ and the force on the block from the ramp? Use a free-body diagram if needed.




3. [10 pt] Three masses are connected by cords, one of which wraps over a pulley of negligible mass and negligible friction on its axle. $m_1$ is on a frictionless plane inclined at angle $\theta = 30°$ (see figure), while the other two hang from the pulley. The three masses are $m_1 = 30\,\mathrm{kg}$, $m_2 = 40\,\mathrm{kg}$, and $m_3 = 20\,\mathrm{kg}$.

(a) When the assembly is released from rest, what is the tension $T_{12}$ in the cord connecting $m_1$ and $m_2$? What about the tension $T_{23}$ in the cord connecting $m_2$ and $m_3$?

(b) How far does $m_1$ move in the first 0.5 second (assuming that $m_1$ starts far enough away from the pulley so it does not reach the pulley)?

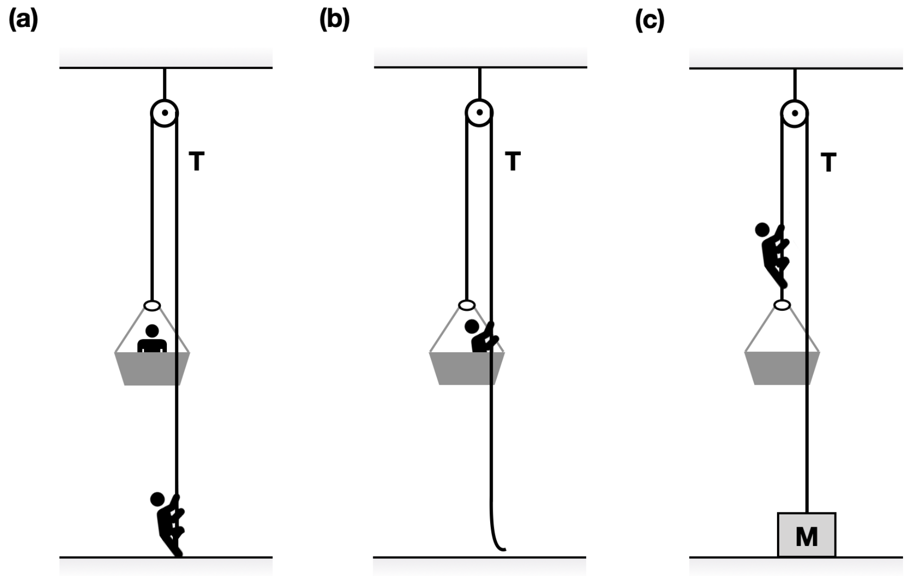



4. [10 pt] A painter is working in a "man cage" hanging from a rope which wraps over a pulley of negligible mass and negligible friction on its axle attached to the ceiling. The man cage can be regarded as massless, while the painter weighs $m = 50\,\mathrm{kg}$.

(a) The rope on the right side of the pulley extends to the ground and is held by a co-worker (see figure). What is the magnitude of force that the co-worker must apply (i.e., tension $T$ in the rope) to keep the painter still? With what force magnitude must the co-worker pull for the painter to rise with a constant speed? With what force magnitude must the co-worker pull for the painter to rise with a constant upward acceleration of $1.2\,\mathrm{m/s^2}$?

(b) The co-worker leaves and now the painter holds the rope (see figure). What force magnitude must he apply to stay still? With what force magnitude must he pull to rise with a constant speed? What if he wishes to rise with a constant upward acceleration of $1.2\,\mathrm{m/s^2}$?

(c) Now a passer-by has noticed that the painter is in a difficult situation and comes to tie the dangling rope to a box of $M = 75\,\mathrm{kg}$, then leaves (see figure). What is the magnitude of the normal force $F_N$ on the box exerted by the ground? What is $F_N$ if the painter climbs up the rope with a constant speed? What if he climbs up the rope with a constant upward acceleration of $1.2\,\mathrm{m/s^2}$ ? Finally, what if he does with a constant upward acceleration of $4.9\,\mathrm{m/s^2}$ ?




5. [10 pt] Ski jumping, one of the most iconic winter sports, consists of three distinct phases: inrun (sliding down the slope), flight, and landing. In each of the three phases, the athlete must control various forces to jump as far down the hill as possible, including the gravity, lift force, air resistance, friction on the slope, normal force, etc. After watching the performance by the current men's record holder at youtube.com/watch?v=RHNPxhpH6qM (or scan the first QR code below with your cellphone camera), describe the balance — or imbalance — of the important forces acting on the jumper in each stage. Two or more paragraphs of 3-4 sentences each is expected to clearly convey your understandings. Use free-body diagrams if desired.



(Note: You are welcomed to study various resources on the web; but surely, your answer must be your own work *in your own words*, and must reference your sources appropriately with a proper citation convention. You may also want to watch a video about the physics behind ski jumping, co-produced by the US National Science Foundation, at youtube.com/watch?v=BDpxSLv89Y8 (or scan the second QR code above with your cellphone camera). To access academic journals off-campus via SNU library's proxy service, see library.snu.ac.kr/using/proxy.)